

Boris Chung

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Mechanical Engineering Graduate | Mechatronics & Systems Integration | Concept to Commissioning

PROFESSIONAL SUMMARY

Mechanical Engineering (Honours) graduate from UQ specialising in mechatronic system design, embedded controls, and hardware integration. Designed and deployed a motorised antenna control system for THIESS Mining, reducing alignment time by over 60% while meeting IP67 and AS3000 standards. Experienced across the full project lifecycle from concept through to on-site commissioning, with two engineering internships in regulated industrial environments. Fluent in English, Cantonese, and Mandarin.

Portfolio: <https://borischung02.github.io/Portfolio/>

TECHNICAL SKILLS

Simulation & Analysis: FEA, STAR-CCM+, Simulink

CAD: Autodesk Inventor, Fusion 360, AutoCAD, Siemens NX

Systems: Hardware integration, electrical system design, AS3000

Manufacturing: CNC machining, assembly, soldering, QC

Programming: Python, MATLAB, C, Java

PROJECTS

2025 - Motorised Antenna Orientation Control – THIESS Mining (Final Year Project)

Designed and engineered a motorised antenna orientation control system for 8m tall masts in collaboration with THIESS Mining. Integrated mechanical design, embedded control, and electrical systems to automate alignment of LTE communication antennas.

- Reduced manual adjustment time from 10 – 15 minutes to under 5 minutes (>60% improvement) through optimized actuation and control logic. Designed for harsh mining environments, meeting IP67 and AS3000 electrical compliance standards.

2024 - Remote-Controlled Tractor Pull System

Designed and developed a remote-controlled tractor prototype capable of generating high-torque traction for load testing applications. Iteratively optimized drivetrain geometry, material selection, and power delivery system to improve pulling performance.

- Achieved 55kg pulling force (91% of design target) through multiple design iterations and mechanical efficiency improvements. Conducted structural and load analysis to validate drivetrain performance under sustained stress conditions.

2022 - Remote-Controlled Mini Fire Truck System

Led the design and development of a remote-controlled fire-response prototype for simulated emergency environments. Integrated mechanical actuation and control systems to enable targeted response to heat sources.

- Designed all 3D-printed components in Fusion 360 and modified store-bought chassis for solid mounting points. Participated in all testing and troubleshooting, extinguishing 4 of 6 simulated fire targets within competition time limits.

WORK EXPERIENCE

June 2025 – August 2025 Engineering Intern, Greater Bay Area Elevator Testing Company, Macau

- Produced 30+ technical tenders and inspection reports for elevator systems, including costing and regulatory documentation
- Conducted on-site mechanical and electrical inspections, identifying compliance issues against local safety standards
- Coordinated multi-stakeholder inspection schedules under tight deadlines in a regulated industrial environment

December 2023 – February 2024 Engineering Intern, TEKOU Const and Eng Company Limited Macau, Macau

- Conducted site visits across public infrastructure sites to assess structural conditions and identify installation locations for a government LED streetscape project
- Researched and shortlisted potential partner companies, preparing presentation materials for stakeholder meetings
- Assisted project coordinator directly, gaining early exposure to construction workflow and professional workplace expectations

EDUCATION

Bachelor of Engineering (Honours)
Mechanical Engineering
The University of Queensland | 2025

LANGUAGES

Fluent in
English | Cantonese | Mandarin
Professional Proficiency in
Japanese

CERTIFICATION & LICENSES

- Temporary Graduate Visa (Subclass 485)
– Full work rights in Australia (valid until Mar 2031)
- Class CA Open Driver License (QLD)
- White Card – Workplace Health and Safety Queensland